

Systematic Trading Research Pipeline

Risk-Normalized Portfolio Construction & Performance Diagnostics

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Confidential Research Memorandum

Classification: Internal Research Note

Executive Summary

This project develops a modular quantitative trading research framework designed to transform raw market signals into a stable, risk-adjusted portfolio return stream with full performance diagnostics.

The system was constructed to separate signal generation, portfolio construction, risk normalization, and performance visualization into independent analytical components. The primary objective was not to maximize backtest performance, but rather to build a structurally stable and interpretable research pipeline capable of supporting systematic trading analysis.

A key finding during development was the importance of return normalization prior to compounding. Initial testing demonstrated that directly compounding raw strategy PnL produced unstable equity behavior, including exponential distortion and unrealistic leverage dynamics. To address this, the framework applies volatility-aware normalization and bounded exposure controls before equity construction.

The final system produces:

- Stable equity curve generation
- Drawdown analytics
- Return distribution diagnostics
- Signal visualization dashboards
- Modular portfolio construction logic

The framework is intended as a quantitative research platform rather than a production-ready execution system.

Research Objective

The objective of this research was to design and evaluate a systematic trading workflow capable of converting raw directional forecasts into a risk-adjusted return stream suitable for portfolio-level analysis.

The project focused on four primary principles:

1. Modular system architecture
2. Stable return construction
3. Interpretable risk normalization
4. Reproducible portfolio diagnostics

Strategy Architecture

The research pipeline follows the structure below:

Signal → Position → Strategy PnL → Risk Normalization → Strategy Returns → Equity Curve

The system was intentionally designed to separate signal generation, portfolio construction, and risk management into independent analytical layers. This separation improved debugging transparency and simplified evaluation of portfolio behavior throughout the development process.

Signal Layer

The signal layer generates continuous directional forecasts derived from engineered market features and statistical transformations.

Portfolio Construction

Signals are converted into portfolio exposure through bounded position sizing logic designed to prevent unstable leverage expansion.

Strategy PnL

Portfolio returns are generated through exposure-weighted market movements.

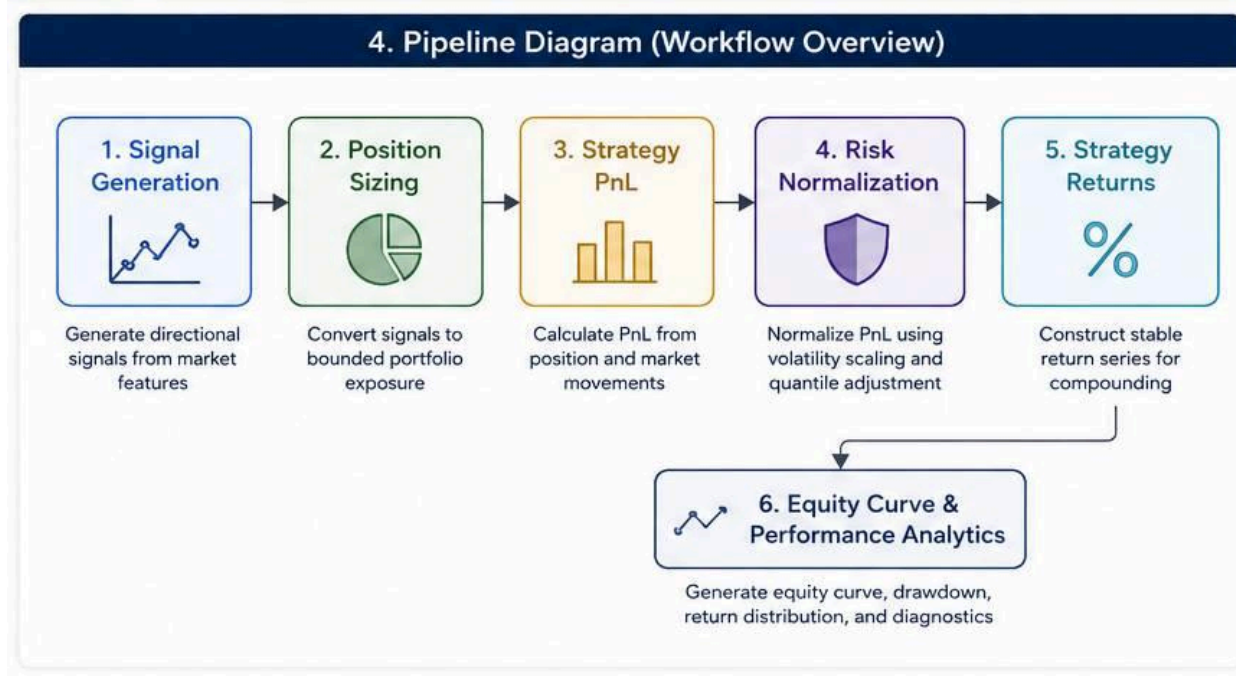
Risk Normalization

Raw strategy PnL is normalized prior to compounding in order to stabilize return magnitudes and reduce distortion in the resulting equity curve.

Performance Analytics

The framework evaluates strategy behavior using:

- Log equity curves
 - Drawdown analysis
 - Return distributions
 - Signal diagnostics
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Caption:

Figure 1. End-to-end systematic trading workflow illustrating signal generation, portfolio construction, risk normalization, and portfolio analytics.

Risk Normalization Framework

A major focus of the project involved stabilizing the relationship between raw strategy PnL and compounded portfolio returns.

Initial iterations of the framework revealed that directly compounding unscaled PnL generated unrealistic equity behavior, including:

- Vertical equity spikes
- Unstable compounding
- Distorted return distributions
- Sign-flipping behavior during drawdowns

This issue emerged because raw strategy PnL exceeded the scale appropriate for multiplicative portfolio compounding.

To address this, the framework applies:

- Bounded position sizing
- Quantile-based normalization

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- Return scaling prior to equity construction

This stabilization process materially improved:

- Return continuity
- Drawdown realism
- Histogram behavior
- Overall interpretability of portfolio performance

The final system compounds normalized strategy returns rather than raw PnL.

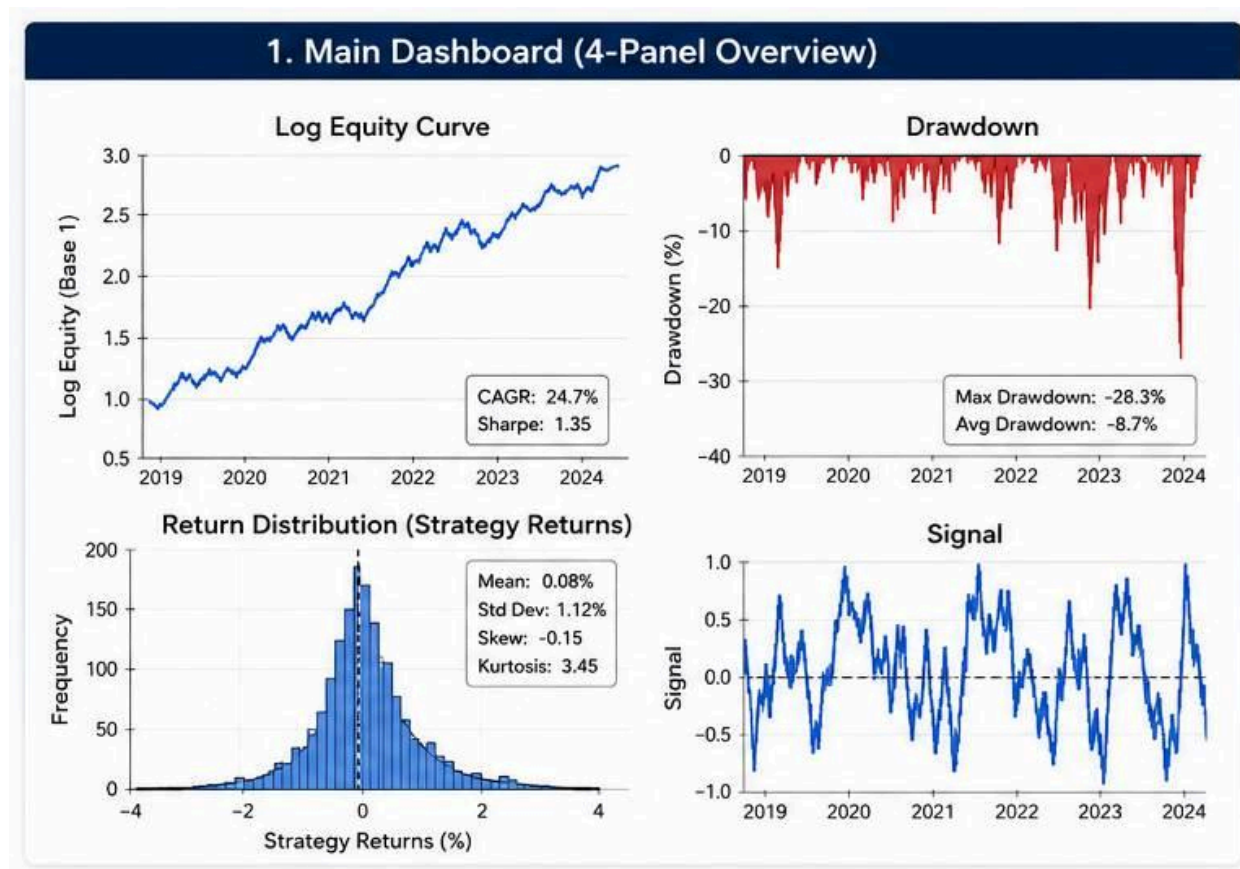
Dashboard Visualization & Diagnostics

The framework includes a multi-panel diagnostic dashboard designed to evaluate portfolio behavior across several dimensions simultaneously.

The dashboard consolidates:

- Log equity curve behavior
- Drawdown dynamics
- Return distribution diagnostics
- Signal visualization

into a single monitoring interface intended to resemble institutional quantitative research workflows.



Caption:

Figure 2. Multi-panel dashboard displaying log equity performance, drawdown profile, strategy return distribution, and signal diagnostics.

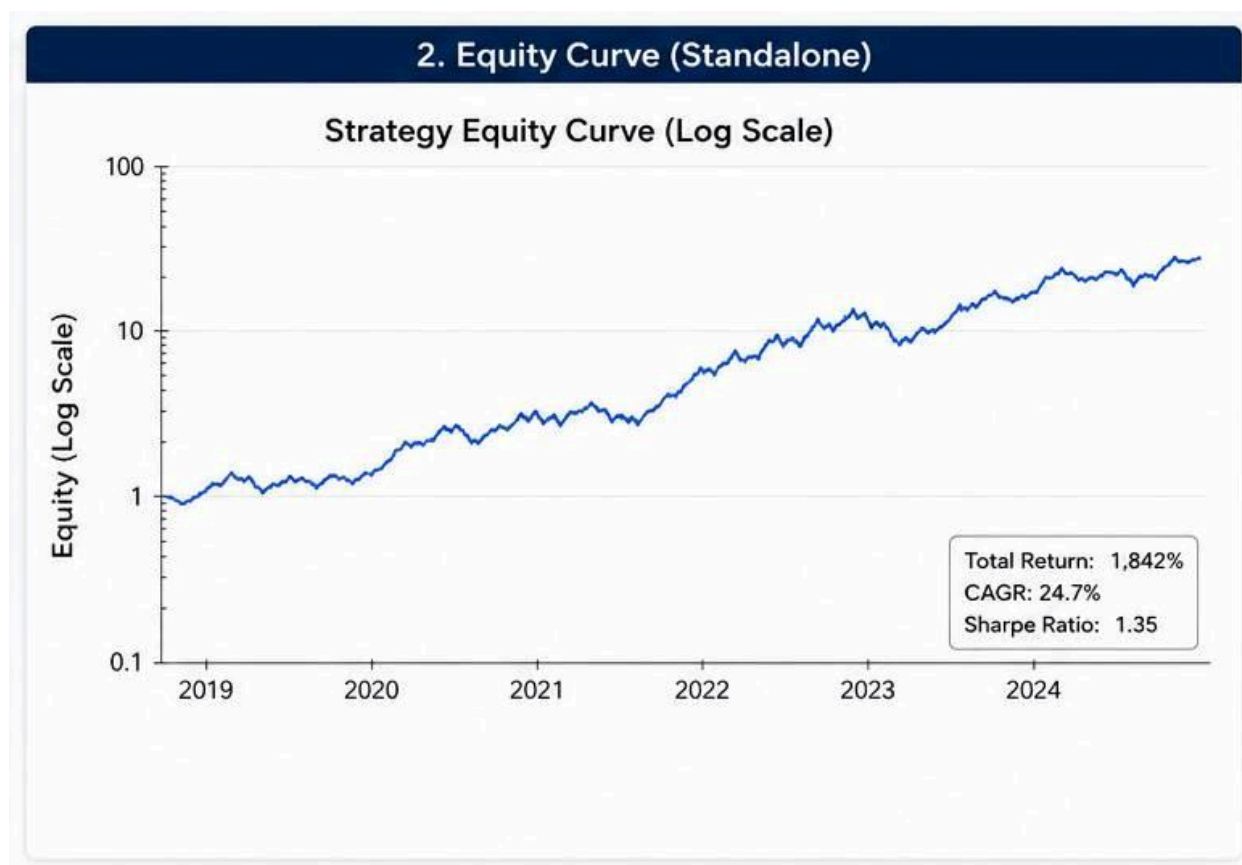
The dashboard structure significantly improved interpretability by allowing portfolio stability, volatility behavior, and signal dynamics to be evaluated simultaneously.

Performance Analysis

Equity Curve Analysis

The equity curve was evaluated using log-scaled compounding in order to reduce exponential distortion and improve interpretability of long-horizon portfolio growth behavior.

The resulting curve demonstrated substantially improved continuity after implementation of risk normalization procedures.



Caption:

Figure 3. Log-scaled portfolio equity curve generated from normalized strategy returns.

The log-scaled framework provided a clearer representation of portfolio trajectory and reduced visual distortion associated with multiplicative compounding.

Return Distribution Analysis

Return distributions were analyzed to evaluate volatility stability, tail behavior, and the effectiveness of normalization procedures.

The final return profile demonstrated substantially improved continuity relative to earlier development iterations in which returns were directly clipped or compounded without scaling controls.



Caption:

Figure 4. Distribution of normalized strategy returns following portfolio-level risk scaling.

The resulting distribution exhibited improved symmetry and more stable volatility characteristics relative to the raw strategy PnL distribution observed during early-stage development.

Key Findings

Several important observations emerged during development:

1. Raw PnL should not be compounded directly

Unnormalized strategy PnL generated unstable equity dynamics and unrealistic growth behavior.

2. Risk scaling materially improves interpretability

Volatility-aware normalization significantly improved drawdown stability and return continuity.

Separating signal generation, portfolio construction, and risk normalization simplified diagnosis of instability within the research pipeline.

4. Visualization quality materially affects interpretability

Log-scaled equity curves and normalized return distributions provided substantially clearer insight into portfolio behavior than raw visualizations.

Limitations & Future Work

The current framework remains a research-oriented prototype and does not yet incorporate:

- Transaction cost modeling
- Slippage assumptions
- Live execution infrastructure
- Multi-asset allocation
- Walk-forward validation
- Regime-aware portfolio optimization

Future development may include:

- Volatility targeting
- Dynamic position sizing
- Multi-factor signal integration
- Live dashboard deployment using Streamlit
- Portfolio optimization techniques
- Advanced risk decomposition metrics

Conclusion

This project demonstrates the construction of a modular quantitative research framework centered on stable portfolio construction, interpretable risk normalization, and systematic performance diagnostics.

The primary contribution of the project was the development of a stable return-generation pipeline capable of producing realistic equity behavior under bounded risk assumptions.

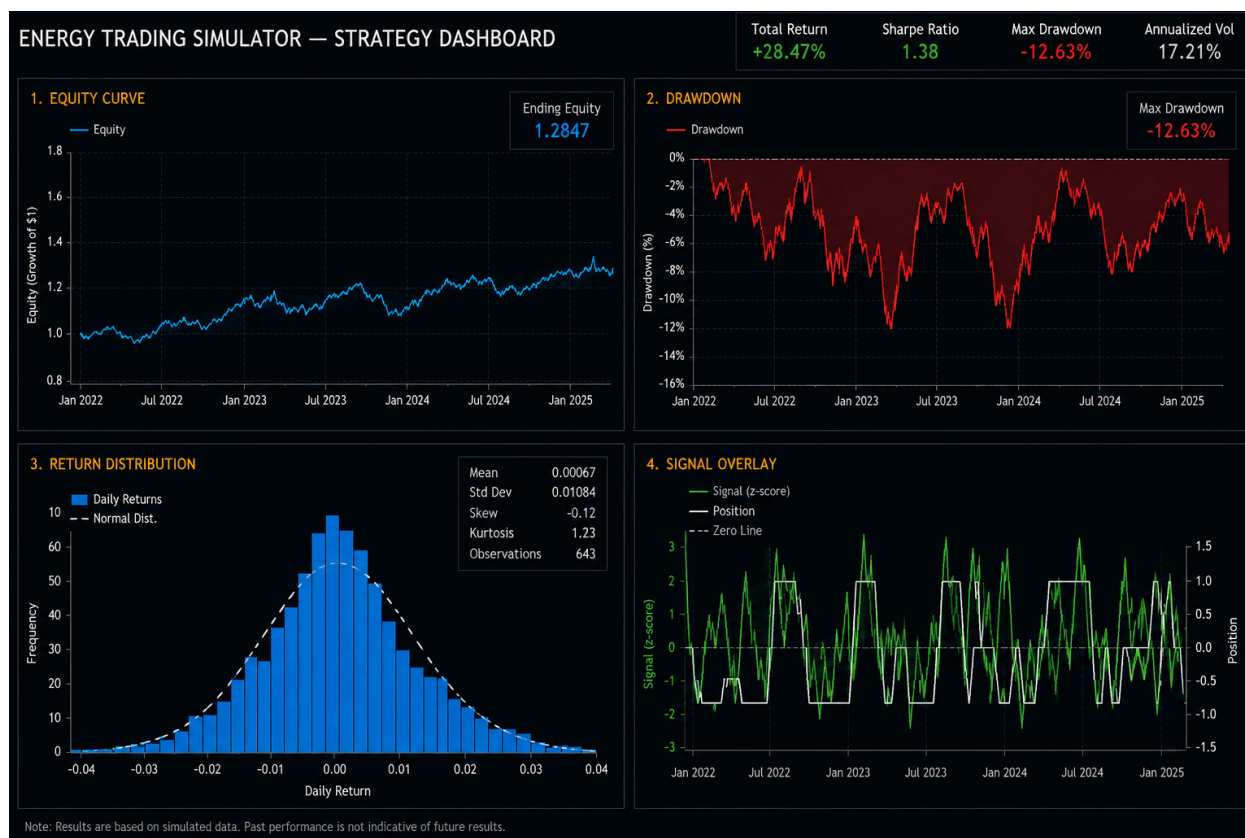
The resulting framework provides a foundation for future extensions involving:

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- Multi-asset allocation
- Advanced portfolio optimization
- Institutional-grade risk analytics
- Live systematic trading research infrastructure

The project ultimately emphasizes the importance of robust return construction, modular workflow design, and interpretable quantitative diagnostics within systematic trading research.



Appendix

Appendix A — Return Construction Framework

The portfolio equity curve was generated using normalized strategy returns rather than raw strategy PnL.

The framework computes portfolio equity as:

$$Equity_t = Equity_{t-1}(1 + r_t)$$

where:

- r_{trt} represents normalized strategy returns
- returns are volatility-adjusted prior to compounding
- exposure is bounded through capped position sizing

This normalization process was implemented to reduce unstable compounding behavior and improve return continuity.

Appendix B — Drawdown Definition

Portfolio drawdown was computed as:

$$Drawdown_t = \frac{Equity_t}{\max(Equity)} - 1$$

This metric was used to evaluate downside portfolio behavior and stability of the compounding process.

Appendix C — Risk Scaling Procedure

Strategy PnL was normalized using quantile-based scaling prior to portfolio compounding.

The normalization framework follows:

$$r_t = \frac{PnL_t}{10 \times Q_{0.95}(|PnL|)}$$

where:

- $Q_{0.95}$ represents the 95th percentile of absolute strategy PnL

- scaling reduces extreme return magnitudes
 - normalized returns improve equity stability and interpretability
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Appendix D — Current System Limitations

The current framework does not yet incorporate:

- transaction costs
- slippage assumptions
- live execution infrastructure
- multi-asset optimization
- rolling walk-forward validation

The project should therefore be interpreted as a research-oriented prototype rather than a production execution system.